

THE SIXTH WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

June 1, 1946

Morning Session

1. Suppose that the function $f(x) = ax^2 + bx + c$, where a, b, c are real constants, satisfies the condition $|f(x)| \leq 1$ for $|x| \leq 1$. Prove that $|f'(x)| \leq 4$ for $|x| \leq 1$. (page 206)

2. If $a(x), b(x), c(x)$, and $d(x)$ are polynomials in x , show that

$$\int_1^x a(x)c(x) \, dx \cdot \int_1^x b(x)d(x) \, dx - \int_1^x a(x)d(x) \, dx \cdot \int_1^x b(x)c(x) \, dx$$

is divisible by $(x - 1)^4$. (page 208)

3. A projectile in flight is observed simultaneously from four radar stations which are situated at the corners of a square of side b . The distances of the projectile from the four stations, taken in order around the square, are found to be R_1, R_2, R_3, R_4 . Show that

$$R_1^2 + R_3^2 = R_2^2 + R_4^2.$$

Show also that the height h of the projectile above the ground is given by

$$h^2 = -\frac{1}{2}b^2 + \frac{1}{4}(R_1^2 + R_2^2 + R_3^2 + R_4^2) \\ - \frac{1}{8b^2}(R_1^4 + R_2^4 + R_3^4 + R_4^4 - 2R_1^2R_3^2 - 2R_2^2R_4^2).$$

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4. Let $g(x)$ be a function that has a continuous first derivative $g'(x)$ for all values of x . Suppose that the following conditions hold for every x : (i) $g(0) = 0$; (ii) $|g'(x)| \leq |g(x)|$. Prove that $g(x)$ vanishes identically. (page 211)

5. Find the smallest volume bounded by the coordinate planes and by a tangent plane to the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1. \quad \text{(page 213)}$$

6. A particle of unit mass moves on a straight line under the action of a force which is a function $f(v)$ of the velocity v of the particle, but the form of this function is not known. A motion is observed, and the distance x covered in time t is found to be connected with t by the formula $x = at + bt^2 + ct^3$, where a, b, c have numerical values determined by observation of the motion. Find the function $f(v)$ for the range of v covered by the experiment.

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Afternoon Session

1. Let K denote the circumference of a circular disc of radius one, and let k denote a circular arc that joins two points a, b on K and lies otherwise in the given circular disc. Suppose that k divides the circular disc into two parts of equal area. Prove that the length of k exceeds 2.

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2. Let A, B be variable points on a parabola P , such that the tangents at A and B are perpendicular to each other. Show that the locus of the centroid of the triangle formed by A, B and the vertex of P is a parabola P_1 . Apply the same process to P_1 , obtaining a parabola P_2 , and repeat the process, obtaining altogether the sequence of parabolas $P, P_1, P_2, \dots, P_n$. If the equation of P is $y^2 = mx$, find the equation of P_n .

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3. In a solid sphere of radius R the density ρ is a function of r , the distance from the center of the sphere. If the magnitude of the gravitational force of attraction due to the sphere at any point inside the sphere is kr^2 , where k is a constant, find ρ as a function of r . Find also the magnitude of the force of attraction at a point outside the sphere at a distance r from the center. (Assume that the magnitude of the force of attraction at a point P due to a thin spherical shell is zero if P is inside the shell, and is m/r^2 if P is outside the shell, m being the mass of the shell, and r the distance of P from the center.)

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4. For each positive integer n , put

$$p_n = (1 + 1/n)^n, P_n = (1 + 1/n)^{n+1}, h_n = \frac{2p_n P_n}{p_n + P_n}.$$

Prove that $h_1 < h_2 < \dots < h_n < \dots$.

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5. Show that the integer next above $(\sqrt{3} + 1)^{2n}$ is divisible by 2^{n+1} .

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6. A particle moves on a circle with center O , starting from rest at a point P and coming to rest again at a point Q , without coming to rest at any intermediate point. Prove that the acceleration vector of the particle does not vanish at any point between P and Q and that, at some point R between P and Q , the acceleration vector points in along the radius RO .

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